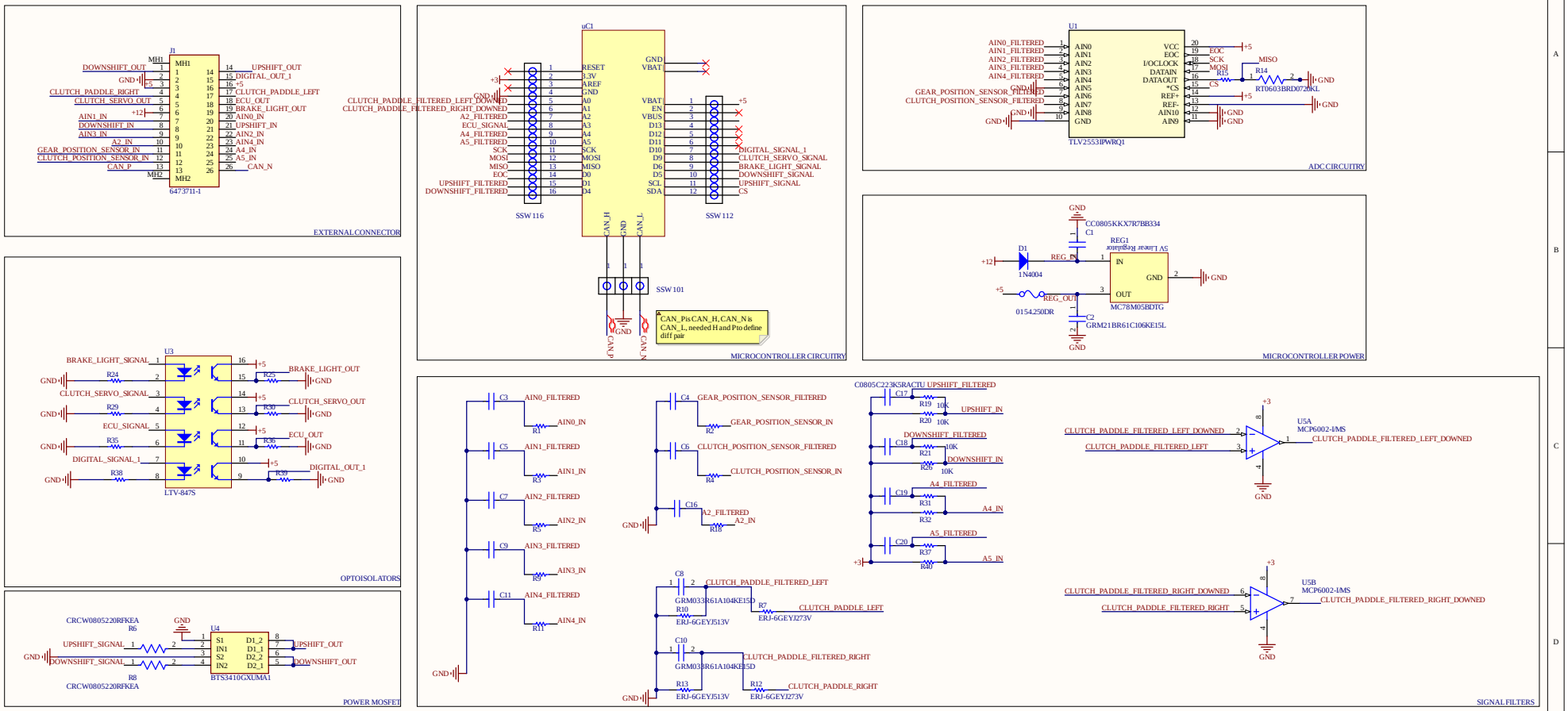

EXTERNAL CONNECTOR
J1
MH1
DOWNSHIFT_OUT
GND
CLUTCH_PADDLE_RIGHT
CLUTCH_SERVO_OUT
AIN1_IN
DOWNSHIFT_IN
AIN3_IN
A2_IN
GEAR_POSITION_SENSOR_IN
CLUTCH_POSITION_SENSOR_IN
CAN_P
MH2
UPSHIFT_OUT
DIGITAL_OUT_1
CLUTCH_PADDLE_LEFT
ECU_OUT
BRAKE_LIGHT_OUT
AIN0_IN
UPSHIFT_IN
AIN2_IN
AIN4_IN
A5_IN
CAN_N
6473711-1
MICROCONTROLLER CIRCUITRY
uC1
RESET
3.3V
AREF
GND
A0
A1
A2
A3
A4
A5
SCK
MOSI
MISO
D0
D1
D4
VBAT
D13
D12
D11
D10
D9
D8
D6
D5
SCL
SDA
CLUTCH_PADDLE_FILTERED_LEFT_DOWNED
CLUTCH_PADDLE_FILTERED_RIGHT_DOWNED
A2_FILTERED
ECU_SIGNAL
A4_FILTERED
A5_FILTERED
UPSHIFT_FILTERED
DOWNSHIFT_FILTERED
DIGITAL_SIGNAL_1
CLUTCH_SERVO_SIGNAL
BRAKE_LIGHT_SIGNAL
DOWNSHIFT_SIGNAL
UPSHIFT_SIGNAL
CS
SSW 116
SSW 112
SSW 101
CAN_H
CAN_L
CAN_P
CAN_N
CAN_Pis CAN_H, CAN_N is CAN_L, neeeded H and P to define diff pair
ADC CIRCUITRY
U1
AIN0_FILTERED
AIN1_FILTERED
AIN2_FILTERED
AIN3_FILTERED
AIN4_FILTERED
GEAR_POSITION_SENSOR_FILTERED
CLUTCH_POSITION_SENSOR_FILTERED
AIN0
AIN1
AIN2
AIN3
AIN4
AIN5
AIN6
AIN7
AIN8
GND
VCC
EOC
I/OCLOCK
DATAIN
DATAOUT
*CS
REF+
REF-
AIN10
AIN9
MISO
MOSI
CS
R14
R15
RT0603BRD0720KL
TLV2553IPWRQ1
MICROCONTROLLER POWER
CC0805KKX7R7BB334
C1
REG1
5V Linear Regulator
D1
1N4004
REG_IN
REG_OUT
0154.250DR
MC78M05BDTG
C2
GRM21BR61C106KE15L
OPTOISOLATORS
U3
BRAKE_LIGHT_SIGNAL
CLUTCH_SERVO_SIGNAL
ECU_SIGNAL
DIGITAL_SIGNAL_1
R24
R29
R35
R38
R25
R30
R36
R39
LTV-847S
POWER MOSFET
CRCW0805220RFKEA
R6
R8
U4
UPSHIFT_SIGNAL
DOWNSHIFT_SIGNAL
S1
IN1
S2
IN2
D1_2
D1_1
D2_2
D2_1
BTS3410GXUMA1
SIGNAL FILTERS
C3
C5
C7
C9
C11
R1
R3
R5
R9
R11
AIN0_FILTERED
AIN1_FILTERED
AIN2_FILTERED
AIN3_FILTERED
AIN4_FILTERED
AIN0_IN
AIN1_IN
AIN2_IN
AIN3_IN
AIN4_IN
C4
C6
C16
R2
R4
R16
GEAR_POSITION_SENSOR_FILTERED
CLUTCH_POSITION_SENSOR_FILTERED
A2_FILTERED
GEAR_POSITION_SENSOR_IN
CLUTCH_POSITION_SENSOR_IN
A2_IN
C0805C223K5RACTU
C17
C18
C19
C20
R19
R21
R31
R37
R20
R26
R32
R40
10K
UPSHIFT_FILTERED
DOWNSHIFT_FILTERED
A4_FILTERED
A5_FILTERED
UPSHIFT_IN
DOWNSHIFT_IN
A4_IN
A5_IN
C8
C10
R10
R13
R7
R12
GRM033R61A104KE15D
ERJ-6GEYJ513V
ERJ-6GEYJ273V
CLUTCH_PADDLE_FILTERED_LEFT
CLUTCH_PADDLE_FILTERED_RIGHT
CLUTCH_PADDLE_LEFT
CLUTCH_PADDLE_RIGHT
U5A
U5B
MCP6002-I/MS
CLUTCH_PADDLE_FILTERED_LEFT_DOWNED
CLUTCH_PADDLE_FILTERED_LEFT
CLUTCH_PADDLE_FILTERED_RIGHT_DOWNED
CLUTCH_PADDLE_FILTERED_RIGHT